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Electronics Design Automation:

The response is so fast it almost mirrors the real thing!

How would you like to design a 15-million-transistor processor using just truth tables and K-maps? Electronic systems are so complex today that electronic design automation (EDA) tools are an absolute necessity. In this interview, we take a look at the advances worth noting in the world of EDA tools. Walden C. Rhines, chairman and CEO of Mentor Graphics, speaks with Dilin Anand from EFY

Q. What would you say are the three features that interest engineers the most, when it comes to emulators?

A. One is the move from traditional in-circuit emulation towards the capability for engineers to do a virtual stimulus. This means that engineers are now able to provide any sort of stimulus for the circuit through software.

The second is to be able to accelerate high-level test benches, whereas the third is for them to be able to do verification of large amounts of embedded software and even application software on the emulator itself.

Q. What is the most exciting recent development in the emulator space for the embedded software developer?

A. One of the things done in the last year has been to improve the performance of software debug from what was typically verifying at about 1MHz to now verifying at about 100MHz. For embedded software developers running something on an emulator means they really cannot tell the difference between whether they are running on an emulator or the real part or a field-programmable gate array board. This is because the response is so fast it almost mirrors the real thing!

Q. What is driving engineers to use EDA tools more often in the automotive sector?

A. In the coming year, the biggest growth will probably come in where embedded software, wiring and other automotive design products are coming into great demand as automotives are becoming more and more electronic. These (automotive companies) have the same design problems as any other company. It is too complex and has to be automated, and automation means EDA.

Q. How have PCB tools evolved over the years?

A. Today, a PCB is not what it used to be. This is because it is not really just limited to PCB but also encompasses system design now.

Q. Is there one significant change that you feel would affect almost every engineer using EDA tools?

A. One change that will affect everyone is the move from the current generation of design for test to the next

generation called Cell-aware test. Traditional automatic test pattern generation (ATPG) assumes gate-level models and stuck-at, transition faults. Cell-aware tests for defects within the standard cell library.

Q. What is your take on EDA's perspective on the Internet of Things (IoT)?

A. The IoT has created a greater interest in multi-die packaging; in analogue and RF for relatively low-complexity designs. It has created interest in hybrid circuits that combine sensors, actuators, micro-electro-mechanical systems (MEMS) and other things with ASIC chips and then package these accordingly. It has affected Big Data processing chips, networking chips and data processing and has pushed the limits on the size of digital chips, so networking companies are making enormous chips to handle all information processing. These would get bigger as you get millions and millions of nodes feeding data and be able to computationally analyse it.

Q. How can engineers at cash-strapped start-ups get access to expensive tools?

A. Twenty years ago, start-ups would offer equity to EDA companies in return for their tools. They later realised that they are better off if they borrowed money and then paid for these tools. At the same time, these companies did not have the funds to equip themselves with the best tools. So we developed programmes where we offer a relatively complete set of design tools at a discounted price during the early years of a company. And as they become more successful, they pay standard prices.

Q. What made you switch from TI to Mentor Graphics, a small company back then?

A. In 1993, I was running the semiconductor business at TI. I was 47 years old, and when I found out that I might have to wait at least 10 years before I could make CEO, I decided to move on. But, I had a non-compete agreement with TI.

When Mentor Graphics called me to tell me they were hunting for a new CEO, I decided to take them up on the offer as I always liked the design and creativity part of semiconductors. ●